🚀 Shine GitOps Project — Assessment Notes

Cluster: shineCluster | **Region:** ap-southeast-2 | **Date:** June 09, 2026

> 🟢 **DONE** = Already completed
> 🟡 **VERIFY** = Run command + take screenshot
> 🔴 **DO** = Must do tomorrow before assessment
> 📸 **SCREENSHOT** = Take this screenshot for submission

🖥️ Your Key Details

Item	Value
EC2 IP	`172.31.44.190`
Cluster	`shineCluster`
Region	`ap-southeast-2`
Namespace	`retail`
ArgoCD URL	`https://a46b04c8ca4d1493cb62e7899ff2bb87-809774554.ap-southeast-2.elb.amazonaws.com`
App URL	`http://a2403bbb14d3946e39dda108c2e48caa-574381716.ap-southeast-2.elb.amazonaws.com:3130`
GitHub Repo	`https://github.com/nanineelapu/Shine-GitOps-Project`

✅ STEP 1 — EKS Cluster Setup

🟢 **DONE**

```bash
```

```bash
# Verify command
kubectl get nodes
```

📷 **SCREENSHOT:** `kubectl get nodes` showing 2 nodes as `Ready`

✅ STEP 2 — Namespaces Created

🟢 **DONE**

```bash
# Verify command
kubectl get ns
```

📷 **SCREENSHOT:** Shows `retail`, `argocd`, `argo-rollouts` namespaces

✅ STEP 3 — ArgoCD Installed

🟢 **DONE**

```bash
# Verify commands
kubectl get pods -n argocd
kubectl get svc -n argocd
```

📷 **SCREENSHOT:** All ArgoCD pods `Running` + service with `EXTERNAL-IP`

✅ STEP 4 — ArgoCD Login

🟢 **DONE**

```bash
# Login command
kubectl port-forward svc/argocd-server -n argocd 8080:443 &
argocd login localhost:8080 --username admin --insecure
```

📷 **SCREENSHOT:** ArgoCD UI showing applications page (browser screenshot)

✅ STEP 5 — Argo Rollouts Installed

🟢 **DONE**

```bash
# Verify command
kubectl get pods -n argo-rollouts
```

📸 **SCREENSHOT:** `argo-rollouts` pod showing `Running`

✅ STEP 6 — Helm Chart Deployed

🟢 **DONE**

```bash
# Verify commands
helm list -n retail
kubectl get pods -n retail
kubectl get svc -n retail
```

📸 **SCREENSHOT:**

- `helm list -n retail` showing `retail-release` as `deployed`
- `kubectl get pods -n retail` showing `1/1 Running` for both pods
- `kubectl get svc -n retail` showing LoadBalancer with EXTERNAL-IP

✅ STEP 7 — ArgoCD Application (GitOps)

🟢 **DONE**

```bash
# Verify commands
argocd app list
argocd app get userprofile-blue-green
```

📸 **SCREENSHOT:**

- ArgoCD UI showing `userprofile-blue-green` app as `Synced` + `Healthy`
- `argocd app list` showing nanineelapu repo

✅ STEP 8 — Blue-Green Deployment

🟢 **DONE** (via Argo Rollouts)

```bash
# Verify command
kubectl get rollout -n retail
kubectl get svc -n retail
```

📸 **SCREENSHOT:** Services showing `userprofile-active-service` and `userprofile-preview-service`

✅ STEP 9 — App Accessible via Browser

🟢 **DONE**

App URL:

```
http://a2403bbb14d3946e39dda108c2e48caa-574381716.ap-southeast-2.elb.amazonaws.com:3130
```

📸 **SCREENSHOT:** App running in browser (open the URL and screenshot it)

✅ STEP 10 — Datadog Monitoring

🟢 **DONE**

```bash
# Verify command
kubectl get pods -n datadog
```

📸 **SCREENSHOT:**

- `kubectl get pods -n datadog` showing pods running
- Datadog UI at `https://app.datadoghq.com` showing infrastructure

✅ STEP 11 — AI Troubleshooting Agent

🟢 **DONE** (Script created)

```bash
# Run the agent
export GROQ_API_KEY="your-key"
python3 ~/ai-troubleshooting-agent.py
```

When running, test with:

```
retail/mongodb-deployment-7bd7f7b99-nmm5x
```

📷 **SCREENSHOT:** Terminal showing AI agent analyzing a pod + response from Groq AI

✅ STEP 12 — GitHub Actions CI/CD

🟢 **DONE**

```bash
# Verify workflow file exists
cat ~/Shine-GitOps-Project/.github/workflows/ci-cd.yml
```

📷 **SCREENSHOT:**

- GitHub repo showing `.github/workflows/ci-cd.yml`
- GitHub Actions tab showing workflow run (green checkmark or in progress)

🔴 STEP 13 — Tomorrow Morning: Restart Cluster

🔴 **DO THIS FIRST TOMORROW**

```bash
```

```
# SSH into EC2 first, then run:
aws eks update-kubeconfig --name shineCluster --region ap-southeast-2

# Verify cluster is alive
kubectl get nodes
kubectl get pods -n retail
kubectl get pods -n argocd
```

🔴 STEP 14 — Tomorrow: Re-login ArgoCD

🔴 DO BEFORE ASSESSMENT

bash

```
# Kill any old port-forward
killall kubectl 2>/dev/null || true

# Port forward ArgoCD
kubectl port-forward svc/argocd-server -n argocd 8080:443 &
sleep 3

# Login
argocd login localhost:8080 --username admin --insecure
```

🔴 STEP 15 — Tomorrow: Run AI Agent for Demo

🔴 DO DURING ASSESSMENT

bash

```
# Set API key
export GROQ_API_KEY="your-groq-key-here"

# Run agent
python3 ~/ai-troubleshooting-agent.py

# When prompted, type:
retail/mongodb-deployment-7bd7f7b99-nmm5x
```

📸 COMPLETE SCREENSHOT CHECKLIST

Take ALL these screenshots before/during assessment:

Cluster & Nodes

- [ ] `kubectl get nodes` — 2 nodes Ready
- [ ] `kubectl get ns` — all namespaces shown

ArgoCD

- [ ] `kubectl get pods -n argocd` — all pods Running
- [ ] `kubectl get svc -n argocd` — LoadBalancer with EXTERNAL-IP
- [ ] ArgoCD UI browser screenshot — app showing Synced + Healthy

Helm & App

- [ ] `helm list -n retail` — retail-release deployed
- [ ] `kubectl get pods -n retail` — 1/1 Running
- [ ] `kubectl get svc -n retail` — services with IPs
- [ ] App in browser — open the LoadBalancer URL

Argo Rollouts

- [ ] `kubectl get pods -n argo-rollouts` — Running

Datadog

- [ ] `kubectl get pods -n datadog` — agents Running
- [ ] Datadog UI screenshot showing infrastructure/metrics

AI Agent

- [ ] Terminal showing AI agent running
- [ ] AI response analyzing a pod

GitHub Actions

- [ ] GitHub Actions tab showing workflow
- [ ] `.github/workflows/ci-cd.yml` in repo

⚡ QUICK COMMANDS FOR ASSESSMENT DAY

```bash
# 1. Connect to cluster
aws eks update-kubeconfig --name shineCluster --region ap-southeast-2

# 2. Check everything is running
kubectl get nodes
kubectl get pods -n retail
kubectl get pods -n argocd
kubectl get pods -n argo-rollouts
kubectl get pods -n datadog
helm list -n retail

# 3. ArgoCD login
kubectl port-forward svc/argocd-server -n argocd 8080:443 &
sleep 3
argocd login localhost:8080 --username admin --insecure
argocd app list

# 4. Run AI Agent
export GROQ_API_KEY="your-key"
python3 ~/ai-troubleshooting-agent.py
```

❌ DO NOT DELETE CLUSTER UNTIL AFTER ASSESSMENT!

```bash
# Delete ONLY AFTER assessment is complete:
eksctl delete cluster --name shineCluster --region ap-southeast-2
```

> ⚠️ This takes 15-20 minutes and is IRREVERSIBLE. Wait until assessment is submitted.

📋 ASSESSMENT TASK MAPPING

Task	What It Is	Status
Task 1	GitHub Actions CI Pipeline	✅ Done

Task	What It Is	Status
Task 2	EKS Cluster Setup	✅ Done
Task 3	ArgoCD GitOps	✅ Done
Task 4	Datadog Monitoring	✅ Done
Task 5	AI Troubleshooting Agent	✅ Done
Task 6	Open Source AIOps	🟡 Verify

Good luck tomorrow! 🚀